

Cloud Computing

Lab-8

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SRN: PES2UG23CS906

Class: 6F

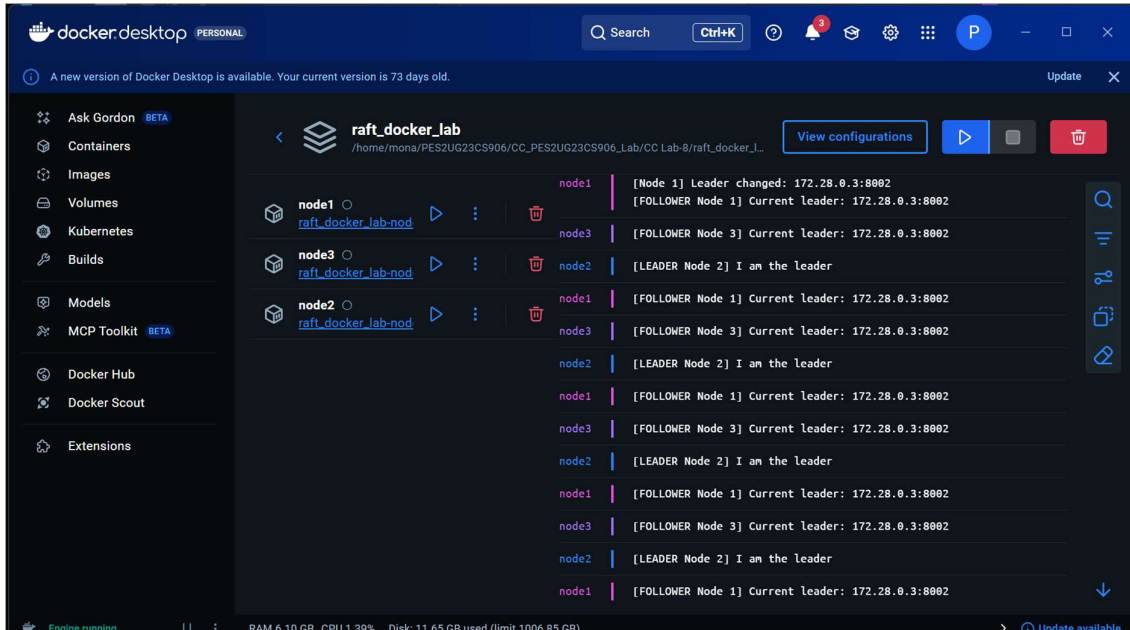
Date: 26/03/2026

1.

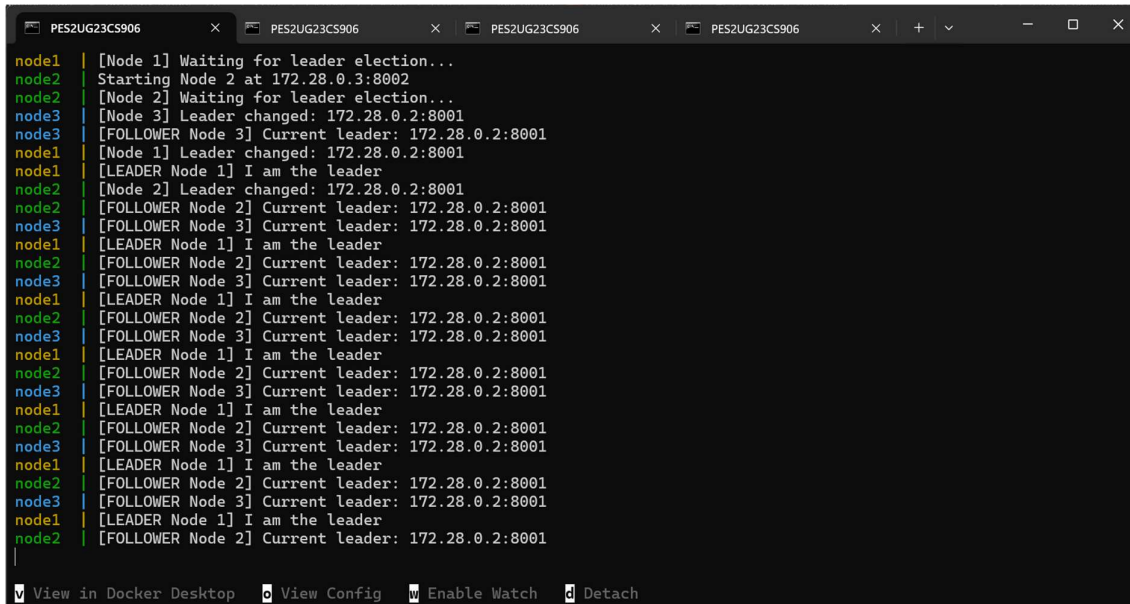
```
'Lab Manual-Student Final.pdf' 'raft_docker_lab (1)'  
mona@MonasLaptop: ~/PES2UG23CS906/CC_PES2UG23CS906_Lab/CC Lab-8$ docker --version  
Docker version 29.1.3, build f52814d  
mona@MonasLaptop: ~/PES2UG23CS906/CC_PES2UG23CS906_Lab/CC Lab-8$ cd raft_docker_lab  
mona@MonasLaptop: ~/PES2UG23CS906/CC_PES2UG23CS906_Lab/CC Lab-8/raft_docker_lab$ dir  
docker-compose.yml Dockerfile node.py requirements.txt  
mona@MonasLaptop: ~/PES2UG23CS906/CC_PES2UG23CS906_Lab/CC Lab-8/raft_docker_lab$ |
```

2.

```
PES2UG23CS906 x mona@MonasLaptop: ~/PES: x mona@MonasLaptop: ~/PES: x + v  
=> => exporting layers  
=> => exporting manifest sha256:8d8490ce6443493fd5295f41602da5444fb9cce081043cc25fa22526f4bcd14a  
=> => exporting config sha256:639601713d27711b10b0ad496640161f77fc3c57155650ca13b680ab68a1a534  
=> => exporting attestation manifest sha256:c153c6a1310aa408fc35f93905db64b21460034c7253de43723036b745  
=> => exporting manifest list sha256:42fd42241fc62dac7ea73efad05ff2c2a07a284ff9cd291059e66adb33fa68a4  
=> => naming to docker.io/library/raft_docker_lab-node1:latest  
=> => unpacking to docker.io/library/raft_docker_lab-node1:latest  
=> [node2] resolving provenance for metadata file  
=> [node1] resolving provenance for metadata file  
=> [node3] resolving provenance for metadata file  
[+] up 7/7  
✔ Image raft_docker_lab-node3 Built  
✔ Image raft_docker_lab-node1 Built  
✔ Image raft_docker_lab-node2 Built  
✔ Network raft_docker_lab-raftnet Created  
✔ Container node1 Created  
✔ Container node2 Created  
✔ Container node3 Created  
Attaching to node1, node2, node3  
node3 | Starting Node 3 at 172.28.0.4:8003  
node3 | [Node 3] Waiting for leader election...  
node2 | Starting Node 2 at 172.28.0.3:8002  
node2 | [Node 2] Waiting for leader election...  
node1 | Starting Node 1 at 172.28.0.2:8001  
node1 | [Node 1] Waiting for leader election...  
node3 | [Node 3] Leader changed: 172.28.0.3:8002  
node3 | [FOLLOWER Node 3] Current leader: 172.28.0.3:8002  
node2 | [Node 2] Leader changed: 172.28.0.3:8002
```



3.




```
PES2UG23CS906 X PES2UG23CS906 X PES2UG23CS906 X PES2UG23CS906 X
mona@MonasLaptop:~/PES2UG23CS906/CC_PES2UG23CS906_Lab/CC Lab-8/raft_docker_lab$ docker logs -f node3
Starting Node 3 at 172.28.0.4:8003
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
```

4.

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PES2UG23CS906 X PES2UG23CS906 X PES2UG23CS906 X PES2UG23CS906 X PES2UG23CS906 X
mona@MonasLaptop:~/PES2UG23CS906/CC_PES2UG23CS906_Lab/CC Lab-8/raft_docker_lab$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
aabc438f4dfb   raft_docker_lab-node1              "python node.py 1"      2 minutes ago Up 2 minutes
b21081eec197   raft_docker_lab-node3              "python node.py 3"      2 minutes ago Up 2 minutes
dc7f29fc0822   raft_docker_lab-node2              "python node.py 2"      2 minutes ago Up 2 minutes
75d8a182dd5e   906_387_390_cc_proj2-gateway       "docker-entrypoint.s..." 45 hours ago  Up 11 minutes (healthy)  0.0.0.
0:4000->4000/tcp, [::]:4000->4000/tcp
415e72010158   906_387_390_cc_proj2-replica2     "docker-entrypoint.s..." 45 hours ago  Up 11 minutes (healthy)  0.0.0.
0:5002->5002/tcp, [::]:5002->5002/tcp
6a30fb24f628   906_387_390_cc_proj2-replic1     "docker-entrypoint.s..." 45 hours ago  Up 11 minutes (healthy)  0.0.0.
0:5001->5001/tcp, [::]:5001->5001/tcp
cddb776d8b5f   906_387_390_cc_proj2-replica3     "docker-entrypoint.s..." 45 hours ago  Up 11 minutes (healthy)  0.0.0.
0:5003->5003/tcp, [::]:5003->5003/tcp
mona@MonasLaptop:~/PES2UG23CS906/CC_PES2UG23CS906_Lab/CC Lab-8/raft_docker_lab$
```

5.

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PES2UG23CS906 x PES2UG23CS906 x PES2UG23CS906 x PES2UG23CS906 x PES2UG23CS906 x + - O X
mona@MonasLaptop:~/PES2UG23CS906/CC_PES2UG23CS906_Lab/CC_Lab-8/raft_docker_lab$ docker network inspect raft_docker_lab-raftnet
[
  {
    "Name": "raft_docker_lab-raftnet",
    "Id": "b6c5a3c4d4e1c331743d0e96d1ae2279a069064ce48dba282268bc99a86e088",
    "Created": "2026-03-26T08:59:34.711671252Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.28.0.0/16",
          "IPRange": "",
          "Gateway": "172.28.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    },
    "Labels": {
      "com.docker.compose.config-hash": "7b955128463677559d4edba13e9fe8f69dd803b2dfc190dd0f2333946fc6a5d0",
      "com.docker.compose.network": "raftnet",
      "com.docker.compose.project": "raft_docker_lab",
      "com.docker.compose.version": "5.0.1"
    }
  }
]
```

```
PES2UG23CS906 x PES2UG23CS906 x PES2UG23CS906 x PES2UG23CS906 x PES2UG23CS906 x + - O X
"com.docker.compose.version": "5.0.1"
},
"Containers": {
  "aabc438f4dfb6e4b9924ac1a85b222a74fd0017c8e9fe4ac04ae167511f6787c": {
    "Name": "node1",
    "EndpointID": "21a7c5d3035814ac3aec4f1ff3f809ab24a573246b23968d1536f5caad3a123a",
    "MacAddress": "aa:8e:b5:20:08:81",
    "IPv4Address": "172.28.0.2/16",
    "IPv6Address": ""
  },
  "b21081eec1971caaf3b4d2b5916fcf1026b96176bc7cd63d1a349f3cb3837748": {
    "Name": "node3",
    "EndpointID": "a3811fe3bb33b7ee1d38aa5470174e5499d92720359c0b5fcffaf81636377d6c",
    "MacAddress": "fa:bc:ac:fc:67:69",
    "IPv4Address": "172.28.0.4/16",
    "IPv6Address": ""
  },
  "dc7f29fc082200a53ff3af6ecc5278d6b979efeeae85666c9e142cfecb58f527": {
    "Name": "node2",
    "EndpointID": "b3766e839520a8d2e692bb52695df508a1a95ef29535aab3fbc6058882848400",
    "MacAddress": "e6:af:3b:81:32:51",
    "IPv4Address": "172.28.0.3/16",
    "IPv6Address": ""
  }
},
"Status": {
  "IPAM": {
    "Subnets": {
      "172.28.0.0/16": {
        "IPsInUse": 6,
        "DynamicIPsAvailable": 65530
      }
    }
  }
}
]
```

6.


```
PES2UG23CS906 X PES2UG23CS906 X PES2UG23CS906 X PES2UG23CS906
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
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[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[Node 2] Leader changed: None
[Node 2] Waiting for leader election...
[Node 2] Leader changed: 172.28.0.3:8002
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
```

```
PES2UG23CS906 x PES2UG23CS906 x PES2UG23CS906 x PES2UG23CS906 x PES2UG23CS906
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
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[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[Node 3] Leader changed: None
[Node 3] Waiting for leader election...
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
```

7.

**raft_docker_lab**
/home/mona/PES2UG23CS906/CC_PES2UG23CS906_Lab/CC Lab-8/raft_docker_l...

View configurations



 node1 ● raft_docker_lab-nod...	 node3 [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
 node3 ● raft_docker_lab-nod...	 node2 [LEADER Node 2] I am the leader
 node2 ● raft_docker_lab-nod...	 node3 [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
	 node2 [LEADER Node 2] I am the leader
	 node1 Starting Node 1 at 172.28.0.2:8001 [Node 1] Waiting for leader election...
	 node3 [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
	 node2 [LEADER Node 2] I am the leader
	 node1 [Node 1] Leader changed: 172.28.0.3:8002 [FOLLOWER Node 1] Current leader: 172.28.0.3:8002
	 node3 [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
	 node2 [LEADER Node 2] I am the leader
	 node1 [FOLLOWER Node 1] Current leader: 172.28.0.3:8002
	 node3 [FOLLOWER Node 3] Current leader: 172.28.0.3:8002


```
PES2UG23CS906 x PES2UG23CS906 x PES2UG23CS906 x PES2UG23CS906 x PES2UG23CS906 x
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
Starting Node 1 at 172.28.0.2:8001
[Node 1] Waiting for leader election...
[Node 1] Leader changed: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
```

ACTIVITY

8.

```
PES2UG23CS906 x PES2UG23CS906 x PES2UG23CS906 x PES2UG23CS906 x PES2UG23CS906 x
mona@MonasLaptop:~/PES2UG23CS906/CC_PES2UG23CS906_Lab/CC Lab-8/raft_docker_lab$ docker ps
CONTAINER ID   IMAGE                                COMMAND                                  CREATED        STATUS        PORTS
691dfc53f824   raft_docker_lab-node1              "python node.py 1"                     20 seconds ago Up 18 seconds
cc40f4633a3a   raft_docker_lab-node3              "python node.py 3"                     20 seconds ago Up 18 seconds
6eae5c8337db   raft_docker_lab-node2              "python node.py 2"                     20 seconds ago Up 18 seconds
7cfffcd4bc26   raft_docker_lab-node4              "python node.py 4"                     20 seconds ago Up 18 seconds
2fcea6b6a7ea   raft_docker_lab-node5              "python node.py 5"                     20 seconds ago Up 18 seconds
75d8a182dd5e   906_387_390_cc_proj2-gateway       "docker-entrypoint.s..."             45 hours ago   Up 19 minutes (healthy)   0.0.0.0:4000->4000
/tcp, [::]:4000->4000/tcp  906_387_390_cc_proj2-gateway-1
415e72010158   906_387_390_cc_proj2-replica2      "docker-entrypoint.s..."             45 hours ago   Up 19 minutes (healthy)   0.0.0.0:5002->5002
/tcp, [::]:5002->5002/tcp  906_387_390_cc_proj2-replica2-1
6a30fb24f628   906_387_390_cc_proj2-replica1      "docker-entrypoint.s..."             45 hours ago   Up 19 minutes (healthy)   0.0.0.0:5001->5001
/tcp, [::]:5001->5001/tcp  906_387_390_cc_proj2-replica1-1
cddb776d8b5f   906_387_390_cc_proj2-replica3      "docker-entrypoint.s..."             45 hours ago   Up 19 minutes (healthy)   0.0.0.0:5003->5003
/tcp, [::]:5003->5003/tcp  906_387_390_cc_proj2-replica3-1
mona@MonasLaptop:~/PES2UG23CS906/CC_PES2UG23CS906_Lab/CC Lab-8/raft_docker_lab$
```


[illegible]

11.

[illegible]

12.

[illegible]

13.

[illegible]

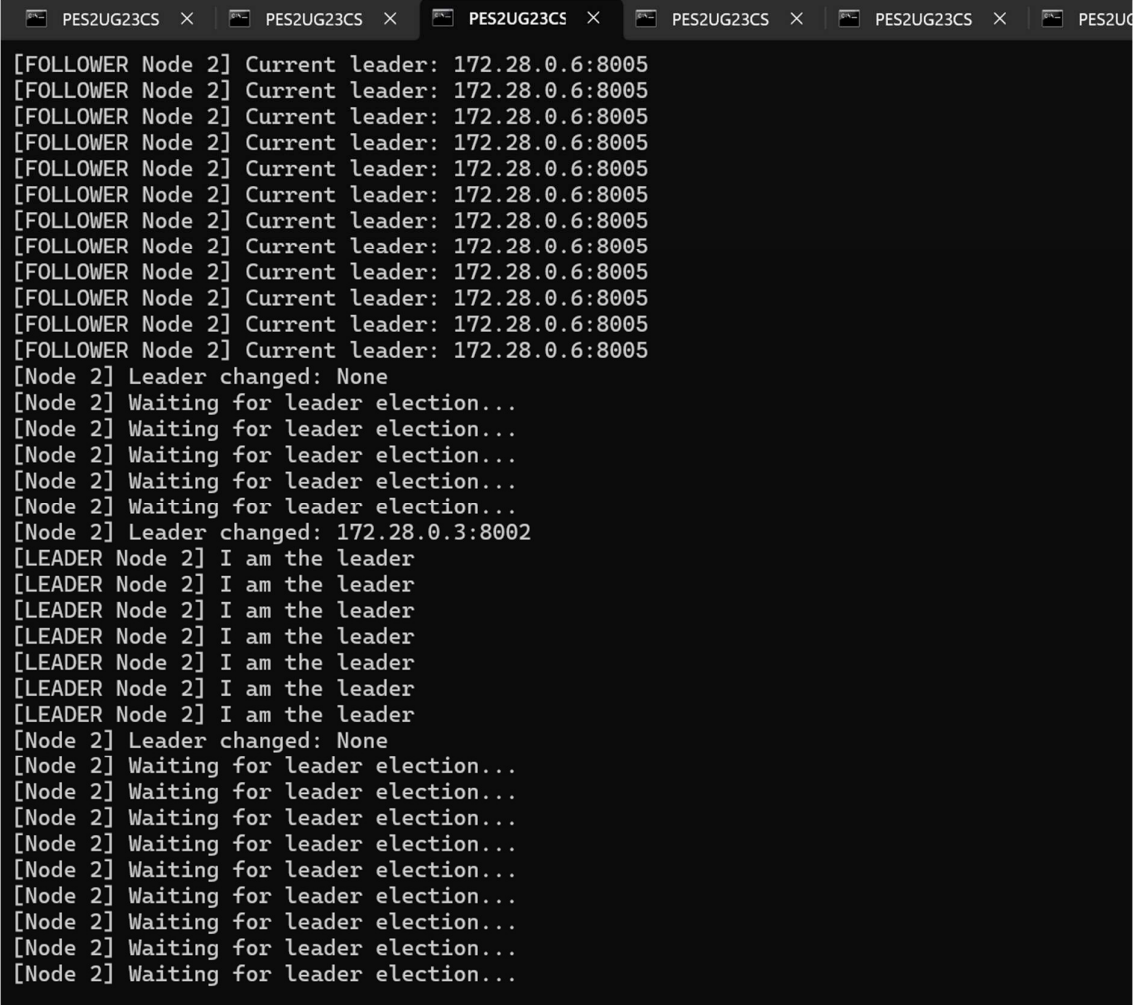
14.

```

mona@MonasLaptop: ~/PES2UG23CS906/CC_PES2UG23CS906_Lab/CC Lab-8/raft_docker_lab$ docker stop node4
node4
mona@MonasLaptop: ~/PES2UG23CS906/CC_PES2UG23CS906_Lab/CC Lab-8/raft_docker_lab$ docker stop node5
node5
mona@MonasLaptop: ~/PES2UG23CS906/CC_PES2UG23CS906_Lab/CC Lab-8/raft_docker_lab$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
691dfc53f824   raft_docker_lab-node1              "python node.py 1"      5 minutes ago Up 5 minutes
cc40f4633a3a   raft_docker_lab-node3              "python node.py 3"      5 minutes ago Up 5 minutes
6eae5c8337db   raft_docker_lab-node2              "python node.py 2"      5 minutes ago Up 5 minutes
75d8a182dd5e   906.387.390_cc_proj2-gateway        "docker-entryptoint.s..." 45 hours ago  Up 24 minutes (healthy)  0.0.0.0:4000->4000/tcp, [::]:4000->4000/tcp
415fe72010158   906.387.390_cc_proj2-replica2        "docker-entryptoint.s..." 45 hours ago  Up 24 minutes (healthy)  0.0.0.0:5002->5002/tcp, [::]:5002->5002/tcp
6a30fb24f628   906.387.390_cc_proj2-replica1        "docker-entryptoint.s..." 45 hours ago  Up 24 minutes (healthy)  0.0.0.0:5001->5001/tcp, [::]:5001->5001/tcp
cddb776d8b5f   906.387.390_cc_proj2-replica3        "docker-entryptoint.s..." 45 hours ago  Up 24 minutes (healthy)  0.0.0.0:5003->5003/tcp, [::]:5003->5003/tcp
mona@MonasLaptop: ~/PES2UG23CS906/CC_PES2UG23CS906_Lab/CC Lab-8/raft_docker_lab$

```


17.



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PES2UG23CS x PES2UG23CS x PES2UG23CS x PES2UG23CS x PES2UG23CS x PES2UG23CS x
[FOLLOWER Node 2] Current leader: 172.28.0.6:8005
[FOLLOWER Node 2] Current leader: 172.28.0.6:8005
[FOLLOWER Node 2] Current leader: 172.28.0.6:8005
[FOLLOWER Node 2] Current leader: 172.28.0.6:8005
[FOLLOWER Node 2] Current leader: 172.28.0.6:8005
[FOLLOWER Node 2] Current leader: 172.28.0.6:8005
[FOLLOWER Node 2] Current leader: 172.28.0.6:8005
[FOLLOWER Node 2] Current leader: 172.28.0.6:8005
[FOLLOWER Node 2] Current leader: 172.28.0.6:8005
[FOLLOWER Node 2] Current leader: 172.28.0.6:8005
[FOLLOWER Node 2] Current leader: 172.28.0.6:8005
[Node 2] Leader changed: None
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Leader changed: 172.28.0.3:8002
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[Node 2] Leader changed: None
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
```

Leader election did not occur when only 2 nodes were active in a 5-node cluster because Raft requires a majority (quorum) of nodes to elect a leader. In a 5-node system, at least 3 nodes must agree for a valid election. With only 2 nodes available, it is impossible to achieve this majority, so no leader can be chosen. This ensures system consistency and prevents incorrect or conflicting decisions in the distributed system.

The Raft distributed cluster was successfully implemented using Docker containers. The system performed leader election and demonstrated fault tolerance when the leader node failed. The experiment was extended to a 5-node cluster to observe quorum behavior. The system continued to function when two nodes were stopped, but became unavailable when three nodes were stopped, proving the majority requirement in Raft.